



EG1113 - C Programming for Engineering Lab

Report #8

String-based Command-line Arguments

Date of the experiment: October 16, 2024

Due date of report: October 18, 2021

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1. Core concepts learned

Summarize the core concepts learned and practiced in this laboratory.

- Setup remote desktop to utilize the assigned Raspberry Pi 400
- Established network file sharing between the Raspberry Pi 400 and laptop utilizing Caja
- Created a hello world program in the C programming language

2. List of equipment and software used

- Raspberry PI 400
- Notepad
- Lenovo Laptop
- Caja File Network Share

3. Relevant web links and other documents:

- Utilized [Geeks For Geeks](#) to understand the string-based command-line arguments.

4. Exercise Make your lab8 work with command line arguments

4.1 Summary

- Modify the program to accept command line arguments that contain the IP addresses of the PI's button board, car IP address, and camera web address.

4.2 Procedures

1. Create a function to extract the parts of a web address

```

9
10 void extract_url_parts(const char *url, char *ip, char *port, char *path)
11 {
12     // Find the start of the IP address after "http://"
13     const char *ip_start = strstr(url, "http://");
14     if (ip_start == NULL) {
15         printf("Invalid URL format.\n");
16         return;
17     }
18     // Skip the "http://" part
19     ip_start += strlen("http://");
20     // Find the end of the IP address, which is marked by ":"
21     const char *ip_end = strchr(ip_start, ':');
22     if (ip_end == NULL) {
23         printf("Invalid URL format: no port found.\n");
24         return;
25     }
26     // Copy the IP address to the output buffer
27     size_t ip_length = ip_end - ip_start;
28     strncpy(ip, ip_start, ip_length);
29     ip[ip_length] = '\0'; // Null-terminate the IP string
30     // Now find the port number, which is between ":" and "/"
31     const char *port_start = ip_end + 1;
32     const char *port_end = strchr(port_start, '/');
33     if (port_end == NULL) {
34         printf("Invalid URL format: no path found.\n");
35         return;
36     }
37     // Copy the port number to the output buffer
38     size_t port_length = port_end - port_start;
39     strncpy(port, port_start, port_length);
40     port[port_length] = '\0'; // Null-terminate the port string
41     // Copy the path (starting from "/")
42     strncpy(path, port_end);
43 }
44

```

2. Modify the main method to accept 3 command line argument

```

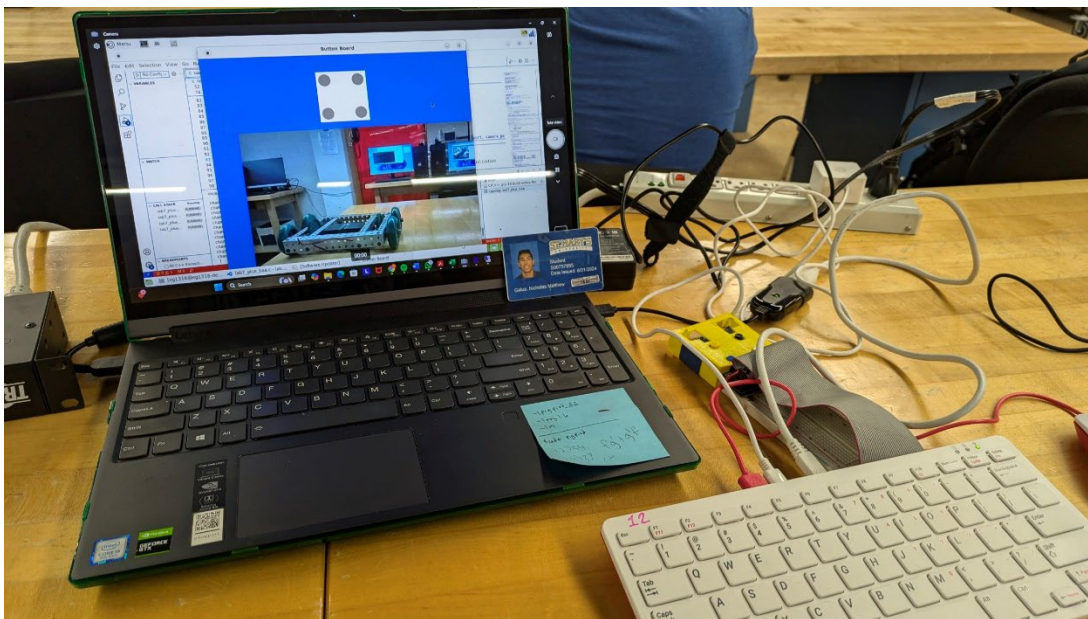
74  if(argc !=1)
75
76  if(argc != 4)
77  {
78  printf("Usage: %s <button_ip car_ip mjpeg_streamer_web_address>\n", argv[0]);
79  exit(_LINE_);
80  }
81
82  // lab 8 code for processing the array of command-line arguments goes here.
83
84  // The code you add will need to change the strings pointed at by:
85  // buttons ip, car ip, camera ip, camera path, and camera port
86  // based on the command line arguments.
87
88  buttons_ip = argv[1]; car_ip = argv[2];
89
90  camera_ip = strdup(argv[3]); camera_port = strdup(argv[3]); camera_path = strdup(argv[3]);
91  extract_url_parts(argv[3], camera_ip, camera_port, camera_path);
92  printf("Here we go - buttons <%s> car <%s> camera <%s><%s><%s>\n", buttons_ip, car_ip, camera_ip, camera_port, camera_path);
93
94

```

5. Exercise Drive a car without using remote-desktop

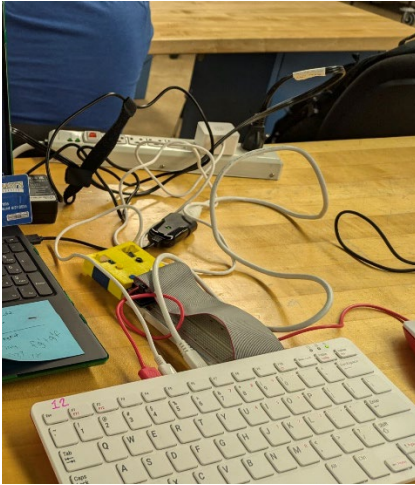
5.1 Summary

- Connect to your pi utilizing the frame grabber and camera application on your laptop
- Lab Verification:

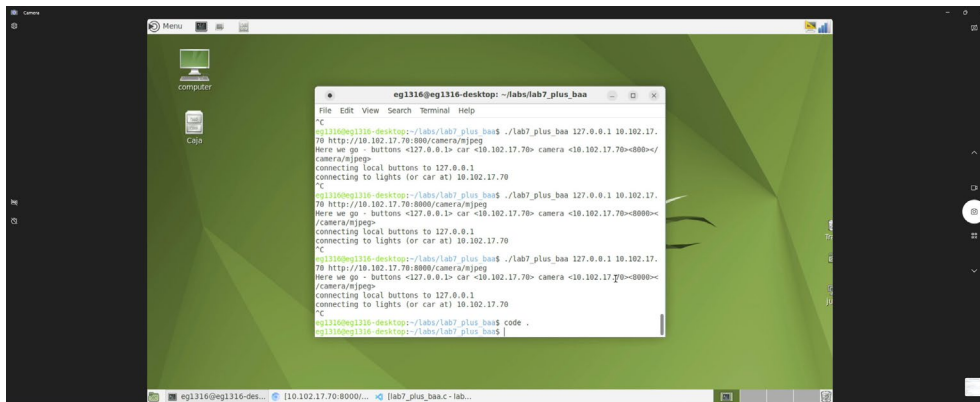


5.2 Procedures

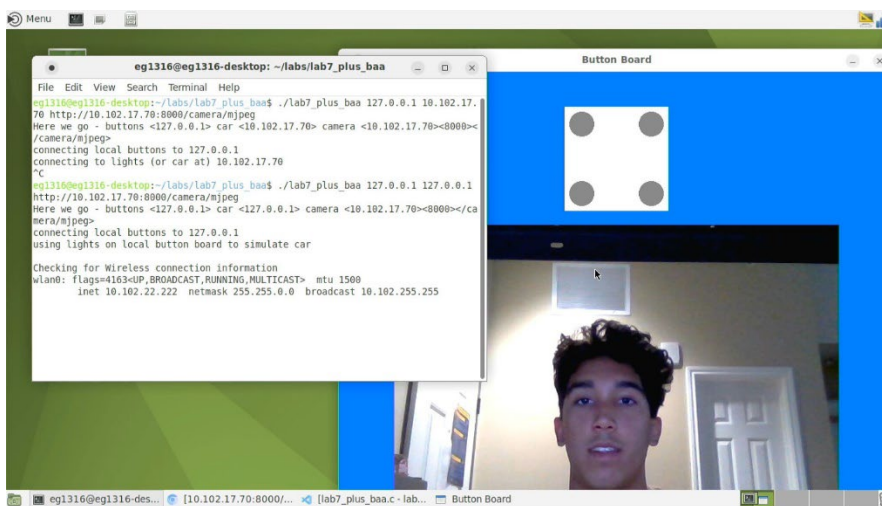
1. Plug HDMI cable to PI and the other end to the frame grabber attached to your laptop
2. Plug in the mouse and power



3. Open camera application to view PI through frame grabber



4. Connect to car utilizing its IP address



6. Daydreaming

- Have the IP address of all the cars saved, so all you need to do is press 1 – 5 corresponding with what car you are using when running program with command line argument
- Utilize a different software to view the PI on windows with the screen grabber to reduce latency or plug PI directly to a display
- Make the program automatically adjust window size in regards to the camera resolution so that it fits within the window

7. Problems experienced and areas of concern

- Was not able to get the command line arguments to work with my code, still debugging it, but was able to it to work with the starter code.

8. Course goals

Rank the level (0=none to 5=a large amount) that this lab covered the each of the course goals by updating the appropriate box in the table below.

Course Goals	Laboratory												
	01	02	03	04	05	06	07	08	09	10	11	12	13
To compile, link, debug and run C programs	4	5	5	5	5	5	5	5					
To use the basic C program structure; variables, constants, and operators	4	4	4	4	4	5	5	5					
To use the repetition constructs such as looping with for, while, and do-while statements	2	3	4	4	4	4	4	4					
To use the selection structures such as if, if/else, switch, conditional expression statement	0	0	3	3	3	3	4	4					
To create program modules using functions (passing data to and returning values from functions)	0	0	0	2	4	4	4	4					
To use arrays and pointers	0	0	0	0	4	3	4	4					
To work with strings and string manipulating functions	2	4	4	1	2	1	2	4					
To perform file, I/O operations	3	4	4	5	5	5	5	5					
To develop structured, modular, and top-down design of software	0	0	0	1	1	4	4	4					